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// Name: Utkarsha Patil
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// Train Ticket Booking System Using Queue

#include <stdio.h>
#define SIZE 5
int q[SIZE];
int wait[SIZE];
int front = -1, rear = -1;
int wcount = 0;
void book(){
   int t;
   printf("Enter ticket number: ");
   scanf("%d", &t);
   if ((rear + 1) % SIZE == front){
      if (wcount == SIZE){
         printf("Waiting list full\n");
      }else{
         wait[wcount] = t;
         wcount++;
         printf("Added to waiting list\n");
      }
   }else{
      if (front == -1){
         front = 0;
      }
      rear = (rear + 1) % SIZE;
      q[rear] = t;
      printf("Ticket booked\n");
   } }
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void cancel(){
   int t, i, j, found = 0;
   printf("Enter ticket number to cancel: ");
   scanf("%d", &t);
   if (front == -1){
      printf("No tickets\n");
      return;
   }
   i = front;
   while (1){
      if (q[i] == t){
         found = 1;
         break;
      }
      if (i == rear){
      break;
      }
      i = (i + 1) % SIZE;
   }
   if (!found){
      printf("Ticket not found\n");
      return;
   }
   while(i != rear){
      q[i] = q[(i + 1) % SIZE];
      i = (i + 1) % SIZE;
   }
   if(front == rear){
   front = rear = -1;
   }else{
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    rear = (rear - 1 + SIZE) % SIZE;
    }
    printf("Ticket cancelled\n");
    if (wcount > 0){
      if (front == -1){
        front = rear = 0;
      }else{
        rear = (rear + 1) % SIZE;
      }
      q[rear] = wait[0];
      printf("Waiting ticket %d confirmed\n", wait[0]);
      for (j = 0; j < wcount - 1; j++){
        wait[j] = wait[j + 1];
        wcount--;
      } } }
void display(){
    int i;
    printf("\nConfirmed Tickets: ");
    if (front != -1){
      i = front;
      while (1){
        printf("%d ", q[i]);
        if (i == rear){
        break;
        }
        i = (i + 1) % SIZE;
      } }
    printf("\nWaiting List: ");
    for (i = 0; i < wcount; i++){
      printf("%d ", wait[i]);
```

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    }
    printf("\n");
}
void main(){
    int op;
    do{
        printf("\n1. Book Ticket\n2. Cancel Ticket\n3. Display Bookings\n4. Exit\n");
        printf("Enter option: ");
        scanf("%d", &op);
        switch (op){
            case 1:
            book();
            break;
            case 2:
            cancel();
            break;
            case 3:
            display();
            break;
            case 4:
            printf("EXITED SUCCESSFULLY\n");
            break;
            default:
            printf("Invalid choice\n");
        }
    }while(op != 4);
}

// Output
1. Book Ticket
```

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 1

Enter ticket number: 101

Ticket booked

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 1

Enter ticket number: 103

Ticket booked

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 1

Enter ticket number: 102

Ticket booked

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 1

Enter ticket number: 104

Ticket booked

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 1

Enter ticket number: 105

Ticket booked

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 3

Confirmed Tickets: 101 103 102 104 105

Waiting List:

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 1

Enter ticket number: 106

Added to waiting list

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 1

Enter ticket number: 107

Added to waiting list

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 3

Confirmed Tickets: 101 103 102 104 105

Waiting List: 106 107

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 2

Enter ticket number to cancel: 102

Ticket cancelled

Waiting ticket 106 confirmed

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 3

Confirmed Tickets: 101 103 104 105 106

Waiting List: 107

1. Book Ticket

2. Cancel Ticket

3. Display Bookings

4. Exit

Enter option: 4

EXITED SUCCESSFULLY